



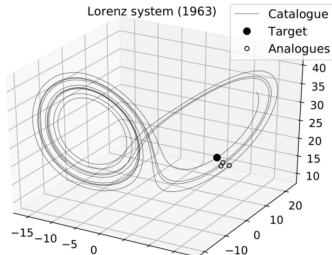
Intercomparison of Analogue Stochastic Forecasts

2nd Stochastic Transport in Upper Ocean Dynamics Annual Workshop
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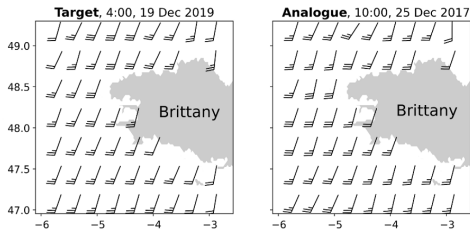
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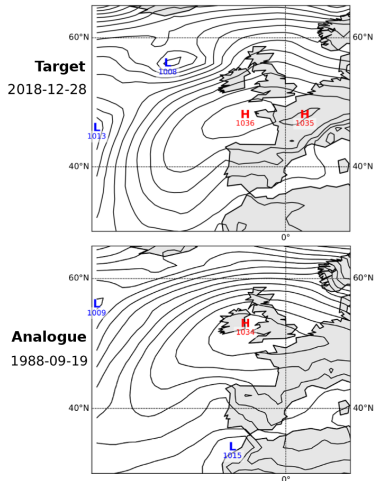
Analogues in idealized and real systems



10m-wind (AROME reanalysis, adapted from Platzer et al. 2021)



Mean Sea-Level Pressure (ERA5 reanalysis)



Topics

Covered

Analogue stochastic forecasting for atmosphere and ocean

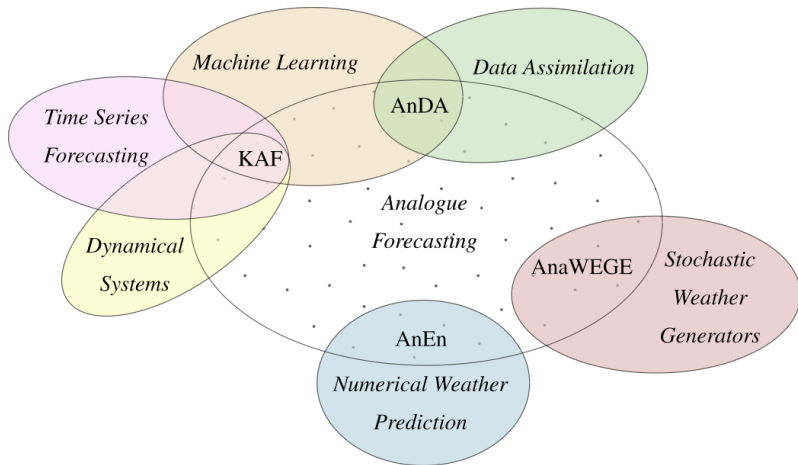
- ▶ Analogue Ensemble (AnEn) (Delle Monache et al., 2013)
- ▶ Analogue Weather Generator (AnaWEGE) (Yiou, 2014)
- ▶ Analogue Data Assimilation (AnDA) (Lguensat et al., 2017)
- ▶ Kernel Analogue Forecasting (KAF) (Alexander and Giannakis, 2020)

Not covered

Other uses of analogues :

- ▶ Downscaling (Zorita and Von Storch, 1999)
- ▶ Extreme event detection and attribution (Jézéquel et al., 2016)
- ▶ Gaining information about the dynamical system
 - Predictability (Lorenz, 1969)
 - Dimension and persistence (Faranda et al., 2017)
 - Recurrence plots (Marwan et al., 2007)
- ▶ ...

Relation with other fields



The other names of analogue methods

- ▶ In machine learning and time series forecasting :
 - “lazy learning” → nothing is computed until a query is given
 - “memory-based learning” → data is stored, not parameters
 - “k-nearest neighbours” (kNN), “local learning”, “local method”, “local model”
- ▶ In statistics :
 - “non-parametric methods”
 - “kernel regression” → a kernel function is used to weight the contribution of each analogue/neighbour

Unifying framework

Dynamical system (unknown) :

$$\Phi^t : \Omega \mapsto \Omega$$

$$\omega_0 \mapsto \omega_t$$

Covariate/predictor/explanatory variable :

$$X : \Omega \mapsto \mathcal{X}$$

$$\omega \mapsto X(\omega)$$

Predictand/response variable :

$$Y : \Omega \mapsto \mathcal{Y}$$

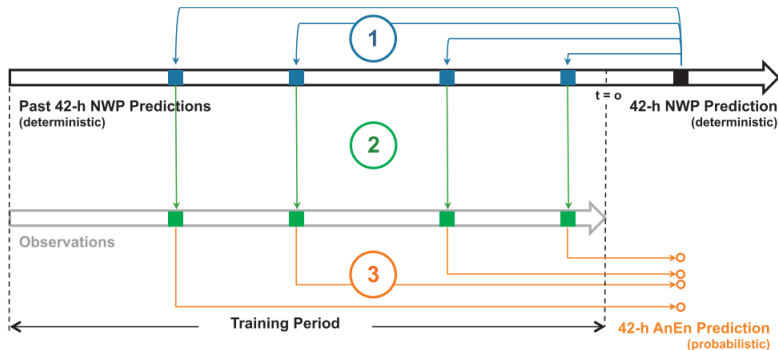
$$\omega \mapsto Y(\omega)$$

Analogue stochastic forecasting :

- ▶ Choose appropriately X
- ▶ Estimate $\mathbb{P}(Y_{t+\tau}|X_t)$ using a catalogue of time-ordered realizations

$$\mathbf{C} = \{x_{t_k}, y_{t_k+\tau}, t_k\}_{k \in [1, n]}$$

Analogue Ensemble (AnEn) : principle



From Delle Monache et al. (2013).

Analogue Ensemble (AnEn) : principle

- Covariate = time-delayed deterministic NWP forecast :

$$X_t = \{\text{NWP}_{t+\tau-\Delta t}, \text{NWP}_{t+\tau}, \text{NWP}_{t+\tau+\Delta t}\}$$

(limited to relevant variables)

- Response = any available observed variable :

$$Y_{t+\tau} = \text{OBS}_{t+\tau}$$

Analogue Ensemble (AnEn) : advantages

- ▶ Fast, simple, tailored to real atmospheric data.
- ▶ Baseline = *already operational & informative* deterministic NWP forecast
- ▶ Outperforms state-of-the-science NWP ensemble.

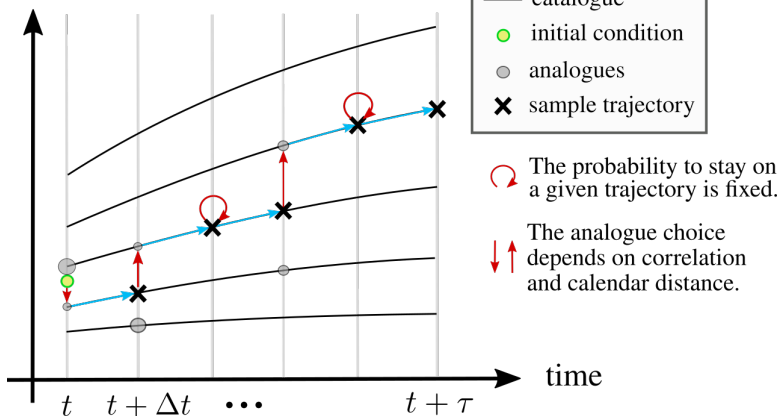
Analogue Ensemble (AnEn) : open questions

- ▶ Hand-tuning (number of members, NWP variables to keep, time-embedding)
- ▶ Dynamical systems formulation ?
- ▶ Test on idealized systems ?
- ▶ Potential risk near bifurcation point (see [supplementary slide](#))

Analogue Weather Generator (AnaWEGE) : principle

Objective = sample *trajectories* starting from a given initial condition
(dynamic mode, see also static mode)

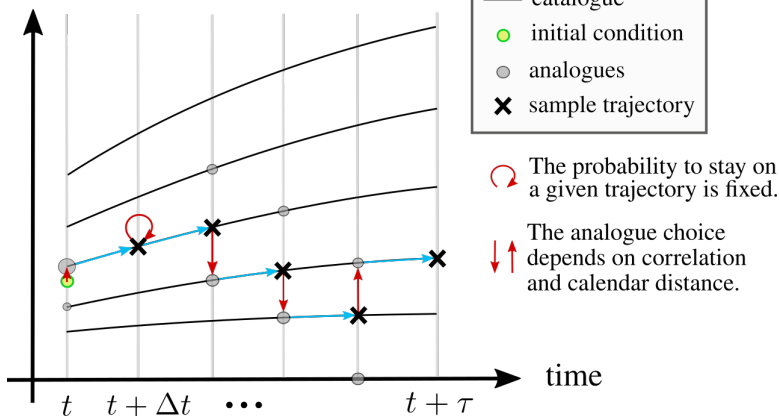
$\mathcal{X}$ = SLP or Z500



Analogue Weather Generator (AnaWEGE) : principle

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$\mathcal{X}$ = SLP or Z500



Analogue Weather Generator (AnaWEGE) : principle

- ▶ Supplementary hypotheses : in the case $\mathcal{X} = \text{SLP}$ or Z500 , the authors assume that $\{X_{n\Delta t}\}_n$ is close to a stationary Markov chain.
- ▶ Then, for any observable Y , observations are generated from the dates of analogue SLP/Z500.

Analogue Weather Generator : advantages

- ▶ Fast, simple, tailored to real atmospheric data.
- ▶ Able to simulate extreme events (hot summer) using importance sampling (Yiou and Jézéquel, 2020).

Analogue Weather Generator : open questions

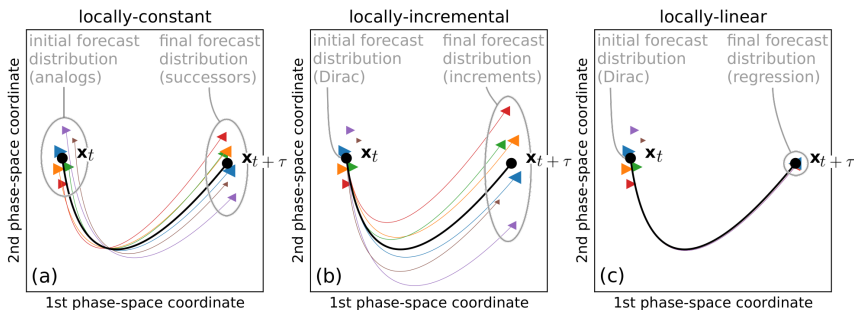
- ▶ Iterated vs. direct analogue forecast ? → time-series forecasting literature (e.g. Bontempi et al., 1999)
- ▶ Hand-tuning (number of analogues, persistence)
- ▶ Dynamical systems formulation ?
- ▶ Test on idealized systems ?

Analog Data Assimilation : forecast (Platzer et al., 2021a)

- $X_{t+\tau} \sim$ discrete distribution (or parametric fit) based on analogs of X_t . Each element is :

$$\text{successor} + \mathbf{S} (\text{target} - \text{analog})$$

$$\mathbf{S} = \begin{cases} \mathbf{0} & \text{(locally-constant)} \\ \mathbf{I} & \text{(locally-incremental)} \\ \text{local regression between analogs \& successors} & \text{(locally-linear)} \end{cases}$$



Analog Data Assimilation : advantages

- ▶ Combine with data assimilation schemes.
- ▶ Fast & simple.
- ▶ Tests on idealized and real systems, dynamical systems formulation.

Remark :

AnDA (LC) $\approx$ AnaWEGE with full persistence .

Analog Data Assimilation : open questions

- ▶ Dynamical systems formulation limited to quasi-Markovian X_t .
- ▶ Forecast error may be overestimated.
- ▶ Hand-tuning (number of analogs, distance function, kernel).
→ use methods from time series forecasting.

Kernel Analog Forecasting : principle

- ▶ Assume w.l.o.g. $\mathcal{Y} = \mathbb{C}$, choose a kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ that measures similarity in $\mathcal{X}$
- ▶ Form a data-based orthonormal basis through orthonormal eigenvectors $\{\phi_{i,n}\}_{1 \leq i \leq n}$ of the matrix $[k(x_{t_i}, x_{t_j})]_{1 \leq i,j \leq n}$. Note l the number of positive eigenvalues $\lambda_{1,n} \geq \dots \geq \lambda_{l,n} > 0$ and $[\mathbf{V}_l]_{i,j} = \lambda_{i,n}^{-1/2} [\phi_{i,n}]_j$ a $l \times n$ projector matrix on the space spanned by $\{\phi_{i,n}\}_{1 \leq i \leq l}$. Note $\mathbf{k}(x) := [k(x, x_{t_1}), \dots, k(x, x_{t_n})]^T$ and $\mathbf{y}_\tau := [y_{t_1+\tau}, \dots, y_{t_n+\tau}]^T$, then the empirical KAF estimator for $\mathbb{E}(Y_{t+\tau} | X_t = x)$ is

$$\hat{f}_{\tau,l,n} = \mathbf{V}_l \mathbf{k}(x) \cdot \mathbf{V}_l \mathbf{y}_\tau$$

⚠ This is not a weighted average (possibly negative coefficients) !

- ▶ Estimate any moment by replacing $\mathbf{y}_\tau$ with it.

Kernel Analog Forecasting : advantages

- ▶ Strong convergence properties. Define Koopman operator $U^\tau \circ Y_t = Y_{t+\tau}$, then :

$$\lim_{l \rightarrow +\infty} \lim_{n \rightarrow +\infty} \hat{f}_{\tau, l, n} = f_\tau ,$$

where f_τ minimizes $\|f_\tau \circ X - U^\tau \circ Y\|$. That is :

$$\lim_{l \rightarrow +\infty} \lim_{n \rightarrow +\infty} \hat{f}_{\tau, l, n}(x_t) = \mathbb{E}(Y_{t+\tau} | X_t = x_t) ,$$

- ▶ Not limited to particular choices of X and Y .
- ▶ Successful applications to ENSO, MJO, idealized deterministic and stochastic dynamical systems...

Kernel Analog Forecasting : open questions

- ▶ Convergence speed ?
- ▶ Computational cost ? $l = \mathcal{O}(10^3)$ and $n = \mathcal{O}(10^4)$ in experiments.
Use of efficient search algorithms ?
- ▶ Gain compared to simple weighted averages ?
- ▶ Applicability to high-dimensional $\mathcal{Y}$?

Recap

- ▶ AnEn and AnaWEGE correspond to specific choices of $\mathcal{X}$.
- ▶ AnaWEGE and AnDA rely on a Markovian X_t .
- ▶ KAF is backed by large data convergence results and valid for any $\mathcal{X}, \mathcal{Y}$.
- ▶ AnEn and AnaWEGE generate ensembles, KAF generates moments.
- ▶ AnDA can generate ensembles and/or moments, and proposes three strategies.

Future research directions

- ▶ Compare all different approaches in a unifying framework and with numerical simulations. Weigh pros and cons (efficiency, computational cost...).
- ▶ Clarify the meaning of AnEn and AnaWEGE using dynamical systems operator theory.
- ▶ Optimize analogue forecasts using results and methods from machine-learning and time series forecasting.
- ▶ Provide a python tool-box (complement to “Flyingpigeon”, web processing service for climate impact and extreme weather event analyses, Hempelmann et al. 2018).

Bibliography (1/3)

- ▶ Delle Monache, L., Eckel, F. A., Rife, D. L., Nagarajan, B., & Searight, K. (2013). Probabilistic weather prediction with an analog ensemble. *Monthly Weather Review*, 141(10), 3498-3516.
- ▶ Yiou, P. (2014). Anawege : a weather generator based on analogues of atmospheric circulation. *Geoscientific Model Development*, 7(2), 531-543.
- ▶ Lguensat, R., Tandeo, P., Ailliot, P., Pulido, M., & Fablet, R. (2017). *The analog data assimilation*. *Monthly Weather Review*, 145(10), 4093-4107.
- ▶ Alexander, R., & Giannakis, D. (2020). Operator-theoretic framework for forecasting nonlinear time series with kernel analog techniques. *Physica D : Nonlinear Phenomena*, 409, 132520.
- ▶ Lorenz, E. N. (1969). Atmospheric predictability as revealed by naturally occurring analogues. *Journal of Atmospheric Sciences*, 26(4), 636-646.
- ▶ Zorita, E., & Von Storch, H. (1999). The analog method as a simple statistical downscaling technique : Comparison with more complicated methods. *Journal of climate*, 12(8), 2474-2489.

Bibliography (2/3)

- ▶ Faranda, D., Messori, G., & Yiou, P. (2017). Dynamical proxies of North Atlantic predictability and extremes. *Scientific reports*, 7(1), 1-10.
- ▶ Jézéquel, A., Yiou, P., & Radanovics, S. (2018). Role of circulation in European heatwaves using flow analogues. *Climate dynamics*, 50(3), 1145-1159.
- ▶ Marwan, N., Romano, M. C., Thiel, M., & Kurths, J. (2007). Recurrence plots for the analysis of complex systems. *Physics reports*, 438(5-6), 237-329.
- ▶ Yiou, P., & Jézéquel, A. (2020). Simulation of extreme heat waves with empirical importance sampling. *Geoscientific Model Development*, 13(2), 763-781.
- ▶ Hempelmann, N., Ehbrecht, C., Alvarez-Castro, C., Brockmann, P., Falk, W., Hoffmann, J., ... & Yiou, P. (2018). Web processing service for climate impact and extreme weather event analyses. Flyingpigeon (Version 1.0). *Computers & Geosciences*, 110, 65-72.
- ▶ Bontempi, G., Birattari, M., & Bersini, H. (1999, June). Local learning for iterated time series prediction. In *ICML* (pp. 32-38).

Bibliography (3/3)

- ▶ Platzter, P., Yiou, P., Naveau, P., Tandeo, P., Zhen, Y., Ailliot, P., & Filipot, J. F. (2021). Using local dynamics to explain analog forecasting of chaotic systems. *Journal of the Atmospheric Sciences*.
- ▶ Platzter, P., Yiou, P., Naveau, P., Filipot, J-F., Thiébaud, M., & Tandeo, P. (2021). Probability distributions for analog-to-target distances, *Journal of the Atmospheric Sciences*.